Ye Wang

Associate Professor | School of Artificial Intelligence
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Employment

Associate Professor, School of Artificial Intelligence, CQUPT — 2026–present
Assistant Professor, School of Computer Science and Technology, CQUPT — 2020–2025
Project Manager (on secondment), China Scholarship Council, Beijing — 2020–2021
Research Scientist, Samsung Research America, United States — 2019

Education

Ph.D. in Computer Engineering, Texas A&M University, United States — 2014–2019
M.S. in Electrical Engineering, The University of Texas at Dallas, United States — 2012–2014
B.Eng. in Microelectronics, Chongqing University of Posts and Telecommunications, China — 2007–2011

Research Overview

My research focuses on **human-centered cognitive AI**, spanning human state understanding, alignment with human preferences, and reasoning under uncertainty. My long-term goal is to develop AI systems that can make appropriate decisions for each individual across different real-world situations. I pursue this goal through three connected directions:

- **Human Understanding** — understanding human states, attributes, individual dynamics, and collective behavior from multimodal and behavioral data. Our work includes multimodal emotion understanding, identity-preserving facial aging, and hierarchical multimodal reasoning for football activity analysis.
- **Human–AI Alignment** — aligning AI systems with diverse human preferences and judgments rather than treating human feedback as a single uniform target. Our recent work studies systematic conflicts among aesthetic attributes and coordinates competing objectives in preference learning.
- **Reasoning under Uncertainty** — enabling AI systems to reason when evidence is incomplete, uncertain, or unreliable. Our work includes uncertainty-aware expert learning, visual-evidence-grounded hallucination correction, and rare safety-critical scenario generation.

Ongoing Research

Building on these foundations, my ongoing research focuses on three questions arising from real-world AI: how to preserve meaningful individual variation, how to reason beyond directly observable evidence, and how to understand interactions among multiple agents.

- **Individual Differences** Modeling how AI can preserve meaningful variation across individuals rather than collapsing them into population-level patterns, enabling context-sensitive preference modeling and personalized trajectories.
- **Reasoning Beyond Visible Evidence** Reasoning about hidden states that cannot be directly observed, using indirect and incomplete evidence to support reliable decisions.
- **Multi-Agent Dynamics** Identifying latent interaction patterns among multiple agents that explain system-level outcomes, including how local interactions and coordination shape collective behavior.

Research Funding

Principal Investigator

2024–2026	**NSFC Young Scientists Fund**	*Controllable Dialogue Generation with Implicit Emotion and Continuous Intent Modeling*
2024–2026	**Chongqing Energy Big Data Center**	*Contract Research Project*

2022–2023	**Chongqing Returned Overseas Scholars Support Program**	*Dialogue Generation with Multi-Granularity Feature Representations*
2021–2024	**Chongqing Municipal Education Commission**	*Interpretable Generative Modeling with Multi-Granularity Temporal Feature Disentanglement*
2020–2025	**CQUPT**	*Neural Architecture Exploration and Optimization with Temporal Features*

Collaborative Projects

2022–2027	**NSFC Innovative Research Group**	*Multi-Granularity Cognitive Computing*
2022–2026	**NSFC Key Program**	*Meso-scale Knowledge Representation for Cognitive Machine Learning*
2022–2025	**National Key R&D Program**	*PB-Scale Multi-Source Health Data Mining*
2020–2024	**NSFC Key Program**	*Concept Embedding for Interpretable Deep Representation Learning*

Research Awards

Spotlight Paper, ICML, Top 2.2% 2026
Gold Reviewer, ICML, 2026
Best Student Paper Award, NCAA, 2026
Best Conference Paper, International Conference on Brain Informatics, 2024
Best Paper Candidate, IEEE ISPCE-ASIA, 2023

Teaching

Teaching Honors

- **2025** — Course Leader, *Data Structures for International Students*; Chongqing First-Class Course.
- **2024** — Award for Outstanding Contribution to International Student Education in Chongqing.
- **2023** — Outstanding Advisor, RoboCom Robot Developer Competition, National Finals.

Student Mentoring

- National Third Prize, Main track, 19th Challenge Cup National College Students' Academic Science Contest.
- Outstanding Undergraduate Thesis of Chongqing.
- National First Prize, Math China Mathematical Modeling Competition.
- National Second Prize, National College Students' Service Outsourcing Innovation Competition.
- National Third Prize, RoboCom Robot Developer Competition.
- Second Prize, Mathematical Contest in Modeling.
- CQUPT Outstanding Undergraduate Thesis Awards.
- National Undergraduate Innovation Training Program.

Courses at CQUPT

- **Natural Language Processing** — *Fall 2024; Fall 2025; Spring 2026*
- **Data Structures for International Students** *Fall 2020; Fall 2022; Spring 2023; Spring 2025*
- **Data Mining** —*Spring 2022; Fall 2022; Fall 2024; Spring 2026*
- **AI Fundamentals and Practice** —*Spring 2025; Fall 2025*
- **Algorithm Analysis and Design for International Students** — *Fall 2023; Fall 2025*
- **Machine Learning** —*Fall 2022; Spring 2024; Fall 2025*
- **Principles of Artificial Intelligence** —*Fall 2020*
- **Big Data Analytics and Mining** —*Fall 2023; Fall 2024*

Teaching Assistant at Texas A&M University

- **ECEN 651: Microprogrammed Control of Digital Systems** — *Fall 2014; Spring 2015.*
- **ECEN 350: Computer Architecture** — *Spring 2015; Fall 2017; Fall 2018; Spring 2019.*
- **ECEN 214: Electrical Circuit Theory** — *Spring 2016.*

Academic Service

Editorial Service

- **Young Editorial Board Member**, CAAI Transactions on Intelligent Systems.
- **Guest Editor**, Special Issue "Explainable AI in Medical Diagnosis: Enhancing Trust and Transparency," *AI*, 2026–2027.
- **Guest Editor**, Special Issue "Advances in Intelligent Transportation Systems Based on Sensor Fusion," *Sensors*, 2024–2025.

Professional Service

- **Committee Member**, CAAI Technical Committee on Granular Computing and Knowledge Discovery.
- **Corresponding Member**, CAAI Technical Committee on Artificial Intelligence Foundations.

Conference Service

- **Session Chair**, Mesoscopic Cognitive Machine Learning Workshop, International Joint Conference on Rough Sets (IJCRS), 2026.
- **Competition Chair**, International Conference on Neural Computing for Advanced Applications (NCAA), 2023–2026.
- **Web Chair**, IEEE International Conference on Medical Artificial Intelligence (MedAI), 2024.

Reviewer

- Conference: ICML (2026); AAAI (2026); COLM (2024–2026); ACM MM (2026); ACL Rolling Review (2025–2026); ECAI (2022–2025); IJCAI (2023–2024).
- Journal: IEEE TPAMI; TIP; TASLP; TCSVT; TKDE; TMM; TAFFC; NN; IF; IPM.

Invited Talks and Media

Invited Talk, "When Attributes Disagree: Gradient Conflict in Image Aesthetic Assessment," China Granular Computing and Knowledge Discovery (CGCKD), Changchun, Jilin — 2026

Invited Talk, "Visual Intelligence through Multi-Granularity Cognitive Computing," CCF@U (No. 1323), Sichuan University — 2025

Invited Talk, "Generative Modeling through Latent-Space Disentanglement," Doctoral Forum, Yan'an University — 2023

Invited Talks, Summer Training Program on Machine Learning, National Center for Applied Mathematics, Chongqing — 2023

Invited Talks, "Generative AI in Education: Opportunities and Challenges," Chongqing Nankai Secondary School — 2023

Media Interview, ChatGPT Is Here: Should We Be Anxious? *Chongqing Daily* — 2023

Representative Publications

[R1] **Ye Wang**, Maocai Dai, Jiang Xie, Xiuli Bi, Fei Tao, Xiao Li, and Hong Yu. "When Attributes Disagree: Gradient Conflict in Image Aesthetic Assessment." *International Conference on Machine Learning (ICML)*, 2026. **Spotlight, Top 2.2%.**

[R2] **Ye Wang**, Pan Sun, Xuyang Zhou, Lifeng Shen, Jiaxu Leng, Guoyin Wang, and Hong Yu*. "Hierarchical Causal Learning for Face Age Synthesis." *IEEE Transactions on Image Processing*, 2026.

[R3] **Ye Wang**, Zixuan Wu, Lifeng Shen, Jiang Xie, Xiaoling Wang, Hong Yu*, and Guoyin Wang. "Mastering the Minority: An Uncertainty-Guided Multi-Expert Framework for Challenging-Tailed Sequence Learning." *IEEE Transactions on Audio, Speech, and Language Processing*, 2026.

[R4] **Ye Wang**, Jiancheng Zhou, Qun Liu*, Feng Hu, and Guoyin Wang. "Visual Evidence-Aware for Object Hallucinations Rectification in LLM-Based Video Captioning." *IEEE Transactions on Circuits and Systems for Video Technology*, 2026.

[R5] **Ye Wang**, Wei Zhang, Ke Liu*, Wei Wu, Feng Hu, Hong Yu, and Guoyin Wang. "Dynamic Emotion-Dependent Network with Relational Subgraph Interaction for Multimodal Emotion Recognition." *IEEE Transactions on Affective Computing*, 2025.

[R6] Xuyang Zhou, **Ye Wang***, Fei Tao, Hong Yu, and Qun Liu. “Hierarchical Chat-Based Strategies with MLLMs for Spatio-Temporal Action Detection.” *Information Processing & Management*, 2025.

[R7] **Ye Wang**, Qingyan Wang, Hong Yu, Jiang Xie, Feng Hu, Xiaoling Wang, and Dajiang Lei*. “GCFA: Generative Class Feature Fusion with Agent Attention for Medical Text Classification.” *Information Fusion*, 2026.

[R8] Tingting Lei, Yifan Zhu, Runxi Zhang, Feng Hu, Hong Yu, and **Ye Wang**. “NLG-Gen: Natural Language-Guided Generation of Long-Tail Critical Scenarios for Autonomous Driving.” *International Conference on Neural Computing for Advanced Applications (NCAA)*, 2026. **Best Student Paper Award.**

[R9] Wenbo Wu, Qingyi Si, Xiurui Pan, **Ye Wang**, and Jie Zhang. “LouisKV: Efficient KV Cache Retrieval for Long Input-Output Sequences.” *International Conference on Learning Representations (ICLR)*, 2026.

[R10] Xinzhe Wang, Fei Tao, Jiang Xie, Hong Yu, and **Ye Wang**. “When Evidence Conflicts: Reliability-Aware Meta-Review Generation.” *2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Findings, 2026.

Books and Monographs

[B1] **Ye Wang**, Hong Yu, and Guoyin Wang. *Multi-Granularity Cognitive Representation*. Science Press, forthcoming, 2026.

Complete Publication List

Journal Papers

[J27] **Ye Wang**, Pan Sun, Xuyang Zhou, Lifeng Shen, Jiaxu Leng, Guoyin Wang, and Hong Yu*. “Hierarchical Causal Learning for Face Age Synthesis.” *IEEE Transactions on Image Processing*, 2026.

[J26] Guangyong He, Li Liu*, Youmin Zhang, **Ye Wang**, Qun Liu, and Guoyin Wang. “Enhancing Graph Neural Network Explainers Using a Distribution Shift Consistency-Guided Generator.” *IEEE Transactions on Knowledge and Data Engineering*, 2026.

[J25] **Ye Wang**, Zixuan Wu, Lifeng Shen, Jiang Xie, Xiaoling Wang, Hong Yu*, and Guoyin Wang. “Mastering the Minority: An Uncertainty-Guided Multi-Expert Framework for Challenging-Tailed Sequence Learning.” *IEEE Transactions on Audio, Speech, and Language Processing*, 2026.

[J24] **Ye Wang**, Jiancheng Zhou, Qun Liu*, Feng Hu, and Guoyin Wang. “Visual Evidence-Aware for Object Hallucinations Rectification in LLM-Based Video Captioning.” *IEEE Transactions on Circuits and Systems for Video Technology*, 2026.

[J23] **Ye Wang**, Qingyan Wang, Hong Yu, Jiang Xie, Feng Hu, Xiaoling Wang, and Dajiang Lei*. “GCFA: Generative Class Feature Fusion with Agent Attention for Medical Text Classification.” *Information Fusion*, 2026.

[J22] Zhuoyi Yu, Guanmeng Xian, Qianmengke Zhao, Hong Yu, Xiao Li, and **Ye Wang***. “Multi-Task Learning with Hierarchical Feature Disentanglement for Low-Exposure Facial Recognition in Smart Cockpits.” *Neural Computing and Applications*, 2026.

[J21] **Ye Wang**, Xinyang Li, Hong Yu, Feng Hu, Guoyin Wang*, and Dajiang Lei*. “Continuous Entity Reasoning for Multi-Turn Medical Dialogue Generation.” *IEEE Transactions on Consumer Electronics*, 2025.

[J20] **Ye Wang**, Qi Wei, Hong Yu, Guoyin Wang, Chunmeng Shi, and Dajiang Lei*. “Cross-Interaction of Chinese Character Structures and Boundary Features for Improving Clinical Named Entity Recognition.” *IEEE Journal of Biomedical and Health Informatics*, 2025.

[J19] Xuyang Zhou, **Ye Wang***, Fei Tao, Hong Yu, and Qun Liu. “Hierarchical Chat-Based Strategies with MLLMs for Spatio-Temporal Action Detection.” *Information Processing & Management*, 2025.

[J18] **Ye Wang**, Wei Zhang, Ke Liu*, Wei Wu, Feng Hu, Hong Yu, and Guoyin Wang. “Dynamic Emotion-Dependent Network with Relational Subgraph Interaction for Multimodal Emotion Recognition.” *IEEE Transactions on Affective Computing*, 2025.

[J17] Yan Xian, Hong Yu, **Ye Wang**, and Guoyin Wang. “Exploring Multi-Granularity Balance Strategy for Class Incremental Learning via Three-Way Granular Computing.” *Brain Informatics*, 2025.

[J16] Chenrui Ma, Yuxin Ding, Haijun Zhang, Jianghong Ma, Zhao Zhang, **Ye Wang**, and Fei Tao. “Fine-Grained Offline Ranking for Short Video Advertisements CTR via Visual- and Audio-Based Features.” *IEEE Transactions on Consumer Electronics*, 2025.

[J15] **Ye Wang**, Zheng Wang, Hong Yu, Guoyin Wang*, and Dajiang Lei*. “The Interactive Fusion of Characters and Lexical Information for Chinese Named Entity Recognition.” *Artificial Intelligence Review*, 2024.

[J14] **Ye Wang**, Qianmengke Zhao, Qun Liu*, Guoyin Wang, Hong Yu, Li Liu, and Jiaxu Leng. “KDDGAN: Knowledge-Guided Explicit Feature Disentanglement for Facial Attribute Editing.” *IEEE Transactions on Consumer Electronics*, 2024.

[J13] Zixuan Wu, **Ye Wang**, Lifeng Shen, Feng Hu, and Hong Yu. “Leveraging Uncertainty for Depth-Aware Hierarchical Text Classification.” *Computers, Materials & Continua*, 2024.

[J12] Hong Yu, Jinxuan Tang, Zhihan Peng, and **Ye Wang**. “Relation Correlations-Aware Graph Convolutional Network with Text-Enhanced for Knowledge Graph Embedding.” *International Journal of Machine Learning and Cybernetics*, 2024.

[J11] **Ye Wang**, Daitianxia Li, Qun Liu, Li Liu, and Guoyin Wang. “Self-Supervised Modal Optimization Transformer for Image Captioning.” *Neural Computing and Applications*, 2024.

[J10] Youmin Zhang, Lei Sun, **Ye Wang***, Qun Liu, and Li Liu. “One-Shot Knowledge Graph Completion Based on Disentangled Representation Learning.” *Neural Computing and Applications*, 2024.

[J9] Kangjia He, Li Liu, Youmin Zhang, **Ye Wang**, Qun Liu, and Guoyin Wang. “Learning Counterfactual Explanation of Graph

Neural Networks via Generative Flow Network." *IEEE Transactions on Artificial Intelligence*, 2024.

[J8] Xu Xiang, Hong Yu, **Ye Wang**, and Guoyin Wang. "Stable Local Interpretable Model-Agnostic Explanations Based on a Variational Autoencoder." *Applied Intelligence*, 2023.

[J7] Jiaxu Leng, Haitao Wang, Xinbo Gao, Yan Zhang, **Ye Wang**, and Mengjingcheng Mo. "Where to Look: Multi-Granularity Occlusion Aware for Video Person Re-Identification." *Neurocomputing*, 2023.

[J6] **Ye Wang**, Jingbo Liao, Hong Yu, Guoyin Wang, Xiaoxia Zhang, and Li Liu. "Interpreting Open-Domain Dialogue Generation by Disentangling Latent Feature Representations." *Neural Computing and Applications*, 2023.

[J5] Xiaoxia Zhang, Lu Chen, **Ye Wang**, and Guoyin Wang. "Improving Incremental Nonnegative Matrix Factorization Method for Recommendations Based on Three-Way Decision Making." *Cognitive Computation*, 2022.

[J4] Li Liu, Yong Du, **Ye Wang**, William K. Cheung, Youmin Zhang, Qun Liu, and Guoyin Wang. "LRP2A: Layer-Wise Relevance Propagation Based Adversarial Attacking for Graph Neural Networks." *Knowledge-Based Systems*, 2022.

[J3] **Ye Wang**, Jingbo Liao, Hong Yu, and Jiaxu Leng. "Semantic-Aware Conditional Variational Autoencoder for One-to-Many Dialogue Generation." *Neural Computing and Applications*, 2022.

[J2] Jiaxu Leng, Yihui Ren, Wen Jiang, Xiaoding Sun, and **Ye Wang**. "Realize Your Surroundings: Exploiting Context Information for Small Object Detection." *Neurocomputing*, 2021.

[J1] **Ye Wang**, Xinxiang Zhang, Mi Lu, Han Wang, and Yoonsuck Choe. "Attention Augmentation with Multi-Residual in Bidirectional LSTM." *Neurocomputing*, 2020.

Conference Papers

[C20] Xinzhe Wang, Fei Tao, Jiang Xie, Hong Yu, and **Ye Wang**. "When Evidence Conflicts: Reliability-Aware Meta-Review Generation." *Findings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

[C19] **Ye Wang**, Maocai Dai, Jiang Xie, Xiuli Bi, Fei Tao, Xiao Li, and Hong Yu. "When Attributes Disagree: Gradient Conflict in Image Aesthetic Assessment." *International Conference on Machine Learning (ICML)*, 2026. **Spotlight, Top 2.2%.**

[C18] Tingting Lei, Yifan Zhu, Runxi Zhang, Feng Hu, Hong Yu, and **Ye Wang**. "NLG-Gen: Natural Language-Guided Generation of Long-Tail Critical Scenarios for Autonomous Driving." *International Conference on Neural Computing for Advanced Applications (NCAA)*, 2026. **Best Student Paper Award.**

[C17] Wenbo Wu, Qingyi Si, Xiurui Pan, **Ye Wang**, and Jie Zhang. "LouisKV: Efficient KV Cache Retrieval for Long Input-Output Sequences." *International Conference on Learning Representations (ICLR)*, 2026.

[C16] Yan Xian, Hong Yu, **Ye Wang**, and Guoyin Wang. "A Novel Class Incremental Learning Method via Multi-Granularity Balance Inspired by Human Granular Cognition Mechanism." *International Conference on Brain Informatics*, 2024. **Best Conference Paper.**

[C15] Pan Sun, Hong Yu, Jiang Xie, Jiaxu Leng, and **Ye Wang**. "Explicit Facial Attribute Disentanglement for Hierarchical Relationships Detection." *International Conference on Neural Computing for Advanced Applications (NCAA)*, 2024.

[C14] Shijie Sun, Hanyue Liu, **Ye Wang**, and Hong Yu. "Lung Cancer Risk Prediction Model Trained with Multi-Source Data." *International Joint Conference on Rough Sets (IJCRS)*, 2024.

[C13] Yali Wang, **Ye Wang**, Hong Yu, and Ke Liu. "Steering Angle Prediction Based on Travelable Regions and Bio-Inspired Neural Circuit Policy." *International Joint Conference on Rough Sets (IJCRS)*, 2024.

[C12] Xinqiang Jiang, Yingnan Geng, Yinzhou Xiong, Fei Tao, and **Ye Wang**. "A Privacy-Aware Framework for Assessing and Recommending Short Video Advertisement." *IEEE International Symposium on Product Compliance Engineering - Asia (ISPCE-ASIA)*, 2023. **Best Paper Candidate.**

[C11] Zhuoyi Yu, **Ye Wang**, and Dajiang Lei. "Research on Chinese Diabetes Question Classification with the Integration of Different BERT Models." *International Conference on Neural Computing for Advanced Applications (NCAA)*, 2023.

[C10] Qingyan Wang, **Ye Wang**, and Dajiang Lei. "SFDA: Chinese Diabetic Text Classification Based on Sentence Feature Level Data Augmentation." *International Conference on Neural Computing for Advanced Applications (NCAA)*, 2023.

[C9] Daitianxia Li, **Ye Wang**, and Qun Liu. "Adaptive Multi-Granularity Aggregation Transformer for Image Captioning." *International Joint Conference on Rough Sets (IJCRS)*, 2023.

[C8] Jiaxu Leng and **Ye Wang**. "RCNet: Recurrent Collaboration Network Guided by Facial Priors for Face Super-Resolution." *IEEE International Conference on Multimedia and Expo (ICME)*, 2022.

[C7] Li Liu, Chongyang Wang, Youmin Zhang, **Ye Wang**, Qun Liu, and Guoyin Wang. "Denoise Network Structure for User Alignment Across Networks via Graph Structure Learning." *International Conference on Data Mining and Big Data (DMBD)*, 2022.

[C6] Jingbo Liao, Hong Yu, Qi Cheng, and **Ye Wang**. "Context-Aware Multi-Feature Fusion for Open-Domain Dialogue Generation." *IEEE International Conference on Industrial Technology (ICIT)*, 2022.

[C5] **Ye Wang**, Han Wang, Xinxiang Zhang, Theodora Chaspari, Yoonsuck Choe, and Mi Lu. "An Attention-Aware Bidirectional Multi-Residual Recurrent Neural Network: A Study on Short-Term Text Classification." *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2019.

[C4] Shan Jin, Jiafan Wang, **Ye Wang**, and Zhengguang Zheng. "A New Method for Sparse Signals Reconstruction in Fusion Center with Incomplete Measurements." *IEEE International Conference on Communications (ICC)*, 2019.

[C3] Xinxiang Zhang, **Ye Wang**, Han Wang, Yue Zhang, Hao Wu, and Mingbo Zhao. "Bend Detection of Bridge Chords in UAV Images via Region-Based Deep Semantic Segmentation Network." *IEEE International Conference on Industrial Informatics (INDIN)*, 2019.

[C2] Han Wang, **Ye Wang**, Xinxiang Zhang, Mi Lu, Yoonsuck Choe, and Jingjing Cao. "English Out-of-Vocabulary Lexical Evaluation Task." *IEEE International Conference on Industrial Informatics (INDIN)*, 2019.

[C1] **Ye Wang**, Yuanjiang Huang, Wu Zheng, Zhi Zhou, Debin Liu, and Mi Lu. "Combining Convolutional Neural Network and Self-Adaptive Algorithm to Defeat Synthetic Multi-Digit Text-Based CAPTCHA." *IEEE International Conference on Industrial Technology (ICIT)*, 2017.